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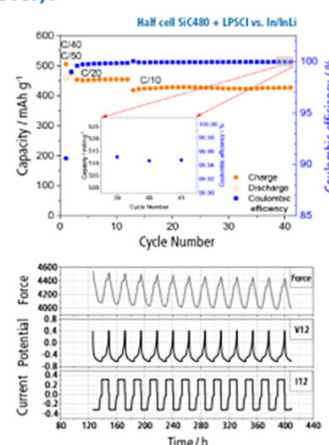
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Voltage-Driven Fast Charging Strategy for Lithium Plating Suppression in Vehicle Batteries via Electrochemical Correlation Mechanism

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During fast charging of lithium-ion batteries in electric vehicles, lithium plating severely degrades performance (e.g., capacity fade, impedance rise) and exacerbates safety risks (e.g., thermal runaway). To address this, we propose a voltage-driven adaptive charging strategy that dynamically regulates current profiles by leveraging the intrinsic correlation between terminal voltage and lithium-plating overpotential. This correlation is first quantified via an electrochemical-thermal-aging (ETA) coupling model. Our analysis reveals that polarization characteristics and plating susceptibility exhibit distinct voltage-range-dependent signatures (independent of ambient temperature or charging current), enabling plating suppression via real-time voltage feedback. Numerical analysis of the current-profile parameters quantitatively elucidated their influence on charging efficiency, guiding the optimization of voltage-threshold-based current regulation. Parameter sweeping confirms that the current profile consistently outperforms conventional constant-current constant-voltage (CC-CV) charging, as well as more advanced MCC and pulse charging strategies, across the entire parameter space. Experimental cycling tests demonstrate that, compared to CC-CV charging, the proposed strategy reduces the capacity loss caused by lithium plating by half and capacity retention rate increased by maximum of 6.8% while maintaining comparable charging time. Critically, this approach eliminates reliance on high-precision onboard state estimators or three-electrode modifications, making it compatible with existing battery management systems (BMS) in electric vehicles and offering a practical pathway to safer fast charging.

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List of Symbols

A	surface area of the cell
A_E	total electrode area
k	ionic conductivity of the electrolyte
k_D	ionic diffusion conductivity of the electrolyte
η_m	activation overpotential of the lithium ion intercalation
η_s	side reaction overpotential
C_e	electrolyte concentration
C_s	ion concentration in solid phase
$C_{s,surf}$	Lithium ion concentration on active material particle surface
$C_{s,max}$	maximum lithium ion concentration of active material particles
D_s	solid diffusivity
D_e	effective ionic diffusivity
E_{act}	activation energy
F	Faraday's constant
i_0	exchange current density
r	material particle radius
t_+	transference number of Li-ion species dissolved in liquid
M	molecular weight
h	heat transfer coefficient of the cell
T_{amb}	ambient temperature
ρ	density of cell
c	specific heat capacity
δ_n	thickness of the negative electrode
δ_p	thickness of the positive electrode
α	transfer coefficients
α_a, α_c	(anodic/cathodic) apparent transfer coefficients
ϵ_e	porosity of electrolyte

σ^{eff}	effective electrical conductivity
ϕ_s, ϕ_e	solid phase/electrolyte potential
U_{ref}	electrode equilibrium potential
U_s^{ref}	side reaction equilibrium potential
U_{ocv}	open circuit voltage
U	terminal voltage
m	cell mass
C_p	specific heat capacity of the cell
Subscripts & Superscripts	
eff	effective value

The large-scale commercialization of pure electric vehicles remains critically challenged by range anxiety and prolonged charging durations, which collectively undermine user confidence and practicality.^{1,2} Enhancing the fast-charging capability of lithium-ion batteries is thus deemed a pivotal technological pathway toward mitigating these barriers.³ Conventional constant current-constant voltage (CC-CV) charging protocols, while widely adopted, often pursue accelerated charging primarily through elevated current rates. Such aggressive current boosting, however, encounters intrinsic limitations: high currents intensify polarization, leading to premature voltage cutoff and consequently restraining actual energy intake. More detrimentally, this practice significantly elevates the risk of lithium plating on the anode surface,⁴⁻⁸ a key degradation mechanism that accelerates capacity loss and poses serious safety hazards.

For lithium-ion batteries of arbitrary specifications currently used in electric vehicles, the onset of lithium plating is reliably indicated when the anode potential (vs Li/Li⁺) falls below 0 V. Monitoring this "plating overpotential" is thus crucial for developing safe fast-charging strategies. Existing approaches to access this key parameter primarily fall into two categories: model-based estimation and direct measurement.

The model-based approach utilizes electrochemical models to infer the overpotential. Klein et al.⁹ formulated a time-optimal charging problem using a simplified electrochemical-thermal model, with the anode overpotential and temperature as constraints.

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Chu et al.¹⁰ further developed a non-destructive, model-based algorithm featuring a dual-loop feedback structure: one loop observes the lithium-plating overpotential in real-time, and the other adjusts the current to keep the lithium-plating overpotential near the preset threshold potential.

The alternative method involves direct measurement using a reference electrode integrated into the cell. For example, Seig et al.¹¹ constructed a three-electrode pouch cell to directly measure the lithium-plating overpotential versus a Li/Li⁺ reference, controlling the charge current to maintain it above 0 V. Waldmann et al.¹² similarly used a reference electrode to study the overpotential's dependence on charging rate, deriving a strategy to avoid plating.

Despite their promise, both methodologies face significant barriers to real-world application in vehicles. Model-based estimation requires highly accurate, parameterized models that remain valid across diverse aging states, posing substantial computational challenges for vehicle-grade microcontrollers.^{13,14} The reference electrode approach, while precise, remains confined to laboratory settings due to cell design modifications and is not feasible for commercial power cells.^{2,15}

This impasse demands easily measurable proxy parameters that reliably indicate plating risk. Li et al.^{16,17} developed a multiscale coupled electrochemical model for the anode that enables accurate calculation of lithium-ion concentration and employs a concentration-based criterion to predict the onset of lithium plating, achieving high prediction accuracy under high C-rate conditions. Subsequently, the onset of lithium plating was identified using the V-Q and dV/dQ curves obtained from charge-discharge experiments on graphite/lithium half cells. However, this method still requires high-fidelity electrochemical models in conjunction with half-cell experiments, rendering it unsuitable for commercial lithium-ion batteries. Given the intrinsic correlation between cell polarization and lithium plating, polarization voltage has been explored as a control metric. Jiang proposed managing end-of-charge polarization to balance speed and safety,¹⁸ while Liu et al. correlated anode polarization with plating using symmetric cells.¹⁹ However, these studies relied on equivalent circuit models lacking deep electrochemical validation. A pivotal advance came from Zoerr et al., who used an electrochemical model to link lithium-plating overpotential to full-cell terminal voltage, enabling a plating-free charging strategy.²⁰ However, this correlation was limited to specific SOC and voltage windows, restricting its utility across the full operational range of vehicle batteries.

To address these limitations, this study establishes a comprehensive, full-range correlation between terminal voltage and lithium-plating overpotential using an electrochemical-thermal-aging (ETA) coupling model. By extending Zoerr's work to the entire SOC and voltage spectrum, we aim to: (1) quantify how polarization voltage and lithium-plating overpotential co-evolve across charging, and evaluate the influence of external factors on both; (2) derive a voltage-driven charging strategy that dynamically adjusts current based on this correlation; and (3) validate its effectiveness in suppressing plating while maintaining charging speed. Unlike model-intensive or hardware-modified approaches, our strategy infers plating risk directly from measurable terminal voltage—eliminating reliance on high-precision estimators or specialized electrodes—and is thus readily integrable with existing battery management systems (BMS) in commercial EVs. This work bridges the gap between electrochemical insights and practical vehicle applications, offering a viable pathway to safer, faster charging.

The remainder of this paper is organized as follows: section 2 presents an overview of the electrochemical-thermal model employed in this study, along with the theoretical basis that were made beforehand. Additionally, second section details a simulation performed in a technology-relevant cell format, aimed at analyzing the variation law of cell polarization voltage and the correlation between lithium-plating overpotential and terminal voltage in the full voltage range. Third section introduces a novel fast charging strategy. Subsequently, the introduced strategy is compared with an equivalent CC-CV charging strategy. Final section concludes with a

summary of the findings from the simulation results discussed earlier.

Basic Principle

Electrochemical mechanism basis.—In this study, a physics-based electrochemical-thermal-aging (ETA) coupled model is employed to establish the underlying mechanistic framework. While the full mathematical formulation of the model is detailed in Ref. 21, a concise summary of the governing equations relevant to this analysis is provided below.

The concentration profile of solid-phase lithium ions is governed by Fick's second law:

$$\frac{\partial C_s}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(D_s r^2 \frac{\partial C_s}{\partial r} \right) \quad [1]$$

The concentration profile of lithium ions in the electrolyte C_e is governed by the following equation:

$$\frac{\partial}{\partial t} \varepsilon_e C_e = \frac{\partial}{\partial x} \left(D_e^{\text{eff}} \frac{\partial C_e}{\partial x} \right) + (1 - t_+) \frac{j_f}{F} \quad [2]$$

The distribution of solid-phase potential within the cell is governed by Ohm's law:

$$\frac{\partial}{\partial x} \left(\sigma^{\text{eff}} \frac{\partial \varphi_s}{\partial x} \right) - j_f = 0 \quad [3]$$

The distribution of liquid-phase potential in the cell is governed by the following equation:

$$\frac{\partial}{\partial x} \left(k^{\text{eff}} \frac{\partial \varphi_e}{\partial x} \right) + \frac{\partial}{\partial x} \left(k_D^{\text{eff}} \frac{\partial}{\partial x} \ln C_e \right) + j_f = 0 \quad [4]$$

Among them, k_D^{eff} is defined as follows:

$$k_D^{\text{eff}} = \frac{2RTk^{\text{eff}}}{F} (t_+ - 1) \quad [5]$$

Since all side reactions during charging occur at the anode, the total volumetric current density at the anode, j_f is the sum of the current densities of the primary and side reactions, i.e.

$$j_f = j_{f,m} + j_{f,s} \quad [6]$$

The volumetric current density of the main reaction, $j_{f,m}$ within the cell is described by the Butler-Volmer equation:

$$j_{f,m} = a_s i_0 \left(\exp \left(\frac{\alpha_a F}{RT} \eta_m \right) - \exp \left(\frac{\alpha_c F}{RT} \eta_m \right) \right) \quad [7]$$

Among them, η_m is the activation overpotential of main reactions in cell, expressed by the following equation:

$$\eta_m = \varphi_s - \varphi_e - U_{\text{ref}} - \frac{R_{\text{SEI}}}{a_s} j_{f,m} \quad [8]$$

Among them, the exchange current density i_0 can be expressed by the following equation:

$$i_0 = i_{0,\text{ref}} (2x^a)(1-x)^a C_e^\alpha \quad [9]$$

Which,

$$x = \frac{C_{s,\text{surf}}}{C_{s,\text{max}}} \quad [10]$$

Side reactions within the cell primarily involve solid electrolyte interphase (SEI) formation and lithium plating processes.

For the lithium plating side reaction, the corresponding side reaction rate equation is as follows:

$$j_{f,Li} = \min \left(0, -a_s i_{0,s} \left(\exp \left(\frac{\alpha_a F}{RT} \eta_{s,Li} \right) - \exp \left(\frac{\alpha_c F}{RT} \eta_{s,Li} \right) \right) \right) \quad [11]$$

Among them, the side reaction overpotential is described by the following equation:

$$\eta_{s,Li} = \varphi_S - \varphi_e - U_s^{ref} - \frac{R_{SEI}}{a_s} j_{f,Li} \quad [12]$$

It can be simplified into the following equation

$$\Delta\varphi = \varphi_S - \varphi_e \quad [13]$$

During high-rate fast charging, lithium plating is the predominant factor contributing to cell aging. Although aging induced by the Solid Electrolyte Interphase (SEI) film growth is comparatively minor, it is unavoidable and can still influence the overall exchange current density. Therefore, to comprehensively characterize battery aging and electrode reaction kinetics the model developed in this study incorporates the side reactions associated with SEI film formation.

The SEI growth reaction rate is described by the Tafel kinetics:²²

$$j_{f,SEI} = -a_s F k_{0,s} c_{EC}^s \exp \left(-\frac{\alpha F}{RT} \eta_{s,SEI} \right) \quad [14]$$

In the equation, $\eta_{s,SEI}$ can be expressed as:

$$\eta_{s,SEI} = \varphi_S - \varphi_e - U_s^{ref} - \frac{R_{SEI}}{a_s} j_{f,SEI} \quad [15]$$

In the formula, $k_{0,s}$ denotes the kinetic rate constant; U_s^{ref} represents the equilibrium potential of the SEI reaction, and c_{EC}^s is the concentration of ethylene carbonate (EC) on the graphite surface, determined according to the principle of mass conservation:

$$-D_{EC} \frac{c_{EC}^s - c_{EC}^0}{\delta_{film}} = -\frac{j_{f,SEI}}{a_s F} \quad [16]$$

In the formula, D_{EC} is the diffusion coefficient of EC, and c_{EC}^0 is the concentration of EC in the bulk electrolyte, δ_{film} is the thickness of the surface facial mask.

The loss of active lithium ions caused by SEI film and lithium plating reaction is calculated using Faraday's law:

$$\frac{\partial c_{SEI}}{\partial t} = -\frac{j_{f,SEI}}{2F} \quad [17]$$

$$\frac{\partial c_{Li}}{\partial t} = -\frac{j_{f,Li}}{2F} \quad [18]$$

Temperature fluctuations have a pronounced impact on the electrochemical behavior of lithium-ion cells during charge–discharge cycles. To accurately capture the internal state of the cell under fast charging conditions, an electrochemical–thermal coupled model is required.

The heat generation within the cell is described by Bernardi's equation:

$$Q = I(U_{ocv} - U) - IT \frac{dU_{ocv}}{dT} \quad [19]$$

The cell's transient thermal behavior is governed by Fourier's law in conjunction with the principle of energy conservation:

$$\rho c \frac{dT}{dt} = \lambda_x \frac{\partial^2 T}{\partial x^2} + \lambda_y \frac{\partial^2 T}{\partial y^2} + \lambda_z \frac{\partial^2 T}{\partial z^2} + Q \quad [20]$$

Among ρ is the cell density, c is the specific heat capacity of the cell, λ is the thermal conductivity of the material, with subscripts x , y , z coordinate axis directions in three-dimensional space; Q is the heat generation rate.

The convective heat dissipation rate of the cell is given by the following equation:

$$q = -hA(T - T_{amb}) \quad [21]$$

Among them, h is the heat transfer coefficient of the cell, A is the outer surface area of the cell, and T_{amb} is the ambient temperature.

The temperature variation of the lithium-ion cell is described by the following equation:

$$mC_p \frac{dT}{dt} = Q - q \quad [22]$$

Where m is the cell mass, and C_p is the specific heat capacity of the cell. The heat transfer coefficient is estimated by analyzing the cell's post-discharge temperature evolution, following the methodology described in Ref. 23. The energy conservation during the cell idle process can be expressed by the following equation:

$$mC_p \frac{dT}{dt} = -hA(T - T_{amb}) \quad [23]$$

Convert the solution of the equation to exponential form:

$$T - T_{amb} = e^{-t/\tau} \quad [24]$$

Among them,

$$\tau = \frac{mC_p}{hA} \quad [25]$$

The heat transfer coefficient of the ternary 811 lithium-ion cell calculated from experimental data at room temperature is 22.77 W/(m²K).²¹

The thermal dependence of the electrochemical properties parameters of lithium-ion cell is described by the Arrhenius equation:

$$\psi = \psi_{ref} \exp \left(\frac{E_{act}}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) \right) \quad [26]$$

Among them, ψ represents a collection of all electrochemical parameters controlled by temperature, ψ_{ref} is their reference value at the reference temperature T_{ref} , and E_{act} represents the activation energy that controls the temperature sensitivity of the parameter. Table A.1 presents the parameter set adopted in the model developed in this study.

Model validation is conducted via rate performance tests. The nominal capacity of the 18650 cylindrical cell is 2.6 Ah, and the positive electrode material is NMC811. The Neware CT-4004 system is used to charge the cell at room temperature. The fully discharged cell is subjected to rate performance tests under various charging rates and measurement techniques that were fully described in our previous work.²¹ The results are shown in Fig. 1, the model provides good prediction for voltage and temperature profiles across various charge rates.

Electrochemical characteristic analysis.—During the initial charging phase, the cell experiences a sharp increase in polarization voltage, leading to a rapid rise in terminal voltage and a significant drop in lithium-plating overpotential. This behavior indicates that

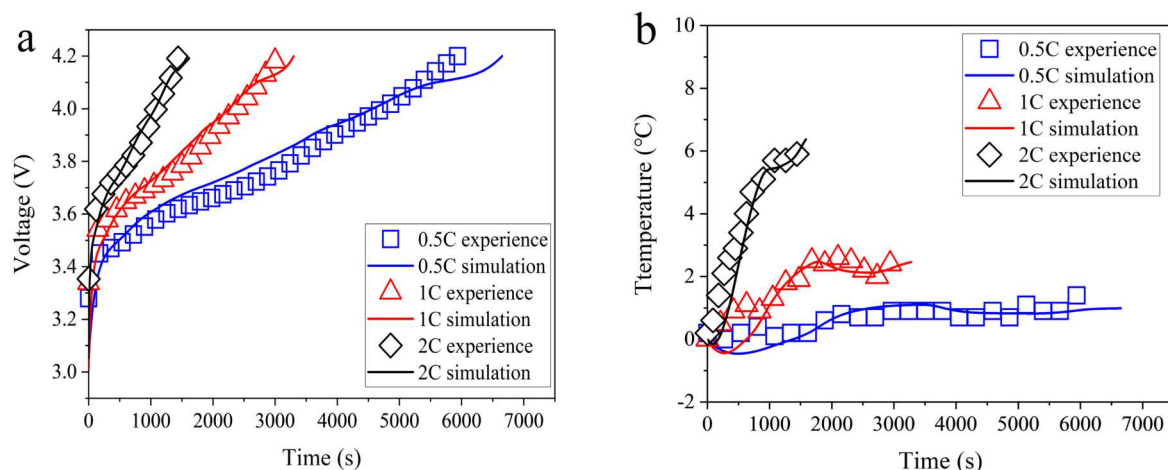


Figure 1. Changes in terminal voltage and temperature of the cell and electrochemical thermal aging coupling model measured at different rates. (a) Voltage distribution. (b) Temperature rise curve.

maintaining a high charging current at this stage is impractical.²⁴ The validated model developed in the previous section was employed to simulate the evolution of polarization voltage and lithium-plating overpotential as the NMC811 cell is charged to 4.2 V under different current rates and operating temperatures. The influence of operational conditions on electrochemical characteristics is systematically evaluated, with the corresponding results illustrated in Figs. 2 and 3.

Figure 2a shows that polarization voltage exhibits a characteristic peak at low SOC, followed by a gradual decline, a trend consistent with the observations reported by Ji et al.²⁴ This behavior is consistent across 3C–5 C charging rates, suggesting that the fundamental behavior of polarization is independent of the applied current magnitude. The initial peak arises because lithium ions have not yet achieved diffusion equilibrium, leading to a steep concentration gradient within the anode active material particles. This gradient exacerbates concentration polarization at the anode, thereby elevating the overall cell polarization. Although the shape of the polarization voltage curve remains similar, the magnitude of the peak is significantly influenced by the charging rate. Higher C-rates intensify the lithium-ion concentration gradient, resulting in a more pronounced polarization voltage peak.

Figure 3a extends the analysis to different temperatures. The polarization peak consistently appears at low SOC regardless of temperature, but its magnitude increases at lower temperatures due to suppressed diffusion and aggravated concentration gradients. These results indicate that polarization pattern is intrinsic and temperature

invariant, while the degree of polarization is strongly temperature dependent.

The lithium-plating overpotential trends shown in Figs. 2b and 3b closely match the polarization evolution: rapid decline of lithium-plating overpotential coincides with the polarization peak, demonstrating a strong mechanistic link, explaining why early-stage charging is most susceptible to lithium plating.

Establishment of full-range voltage-overpotential correlation.—Since the anode lithium-plating overpotential is a reliable indicator of lithium plating risk—with plating initiation confirmed when the lithium-plating overpotential falls below 0 V.²⁰ To develop a voltage-driven fast-charging strategy, it is essential to establish a reliable mapping between terminal voltage and lithium-plating overpotential. Prior studies reported an approximate linear correlation but limited to narrow SOC windows, restricting practical applicability.^{15,18} Therefore, by simulating constant-current charging processes from a fully discharged state to 4.2 V under various current rates and temperatures to analyze and characterize the relationship between the anode lithium-plating overpotential (hereafter referred to as lithium-plating overpotential) and the terminal voltage across the full operational range of the cell. The resulting correlation is illustrated in Fig. 4, which depicts the relationship between the lithium-plating overpotential at the anode-separator interface and the terminal voltage.

The fully discharged cell is charged to 4.2 V at charging rates of 3 C, 4 C, and 5 C in an isothermal environment at 25 °C. Figure 4a

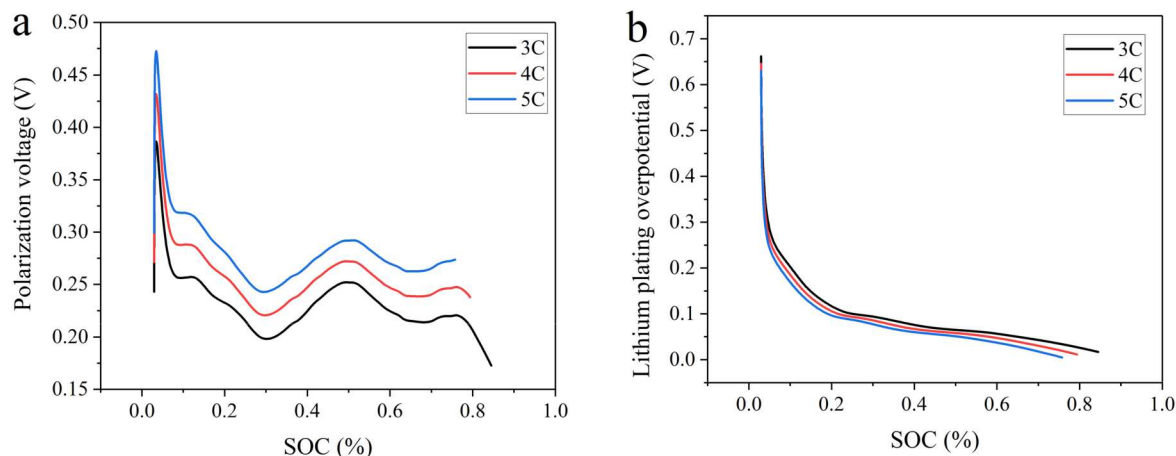


Figure 2. Polarization voltage and lithium-plating overpotential of the cell during charging at 3 C, 4 C, and 5 C rates at 25 °C as a function of SOC. (a) Polarization voltage curve. (b) Lithium-plating overpotential curve at the interface between the anode and the separator.

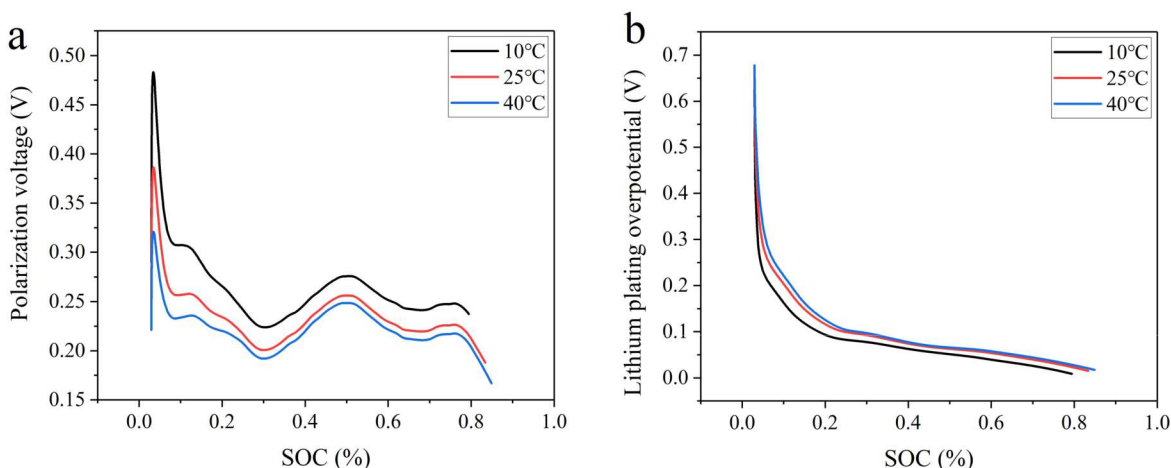


Figure 3. The polarization voltage and lithium-plating overpotential of the cell during charging with a 3 C rate at 10 °C, 25 °C, and 40 °C as a function of SOC. (a) Polarization voltage curve. (b) Lithium-plating overpotential curve at the interface between the anode and the separator.

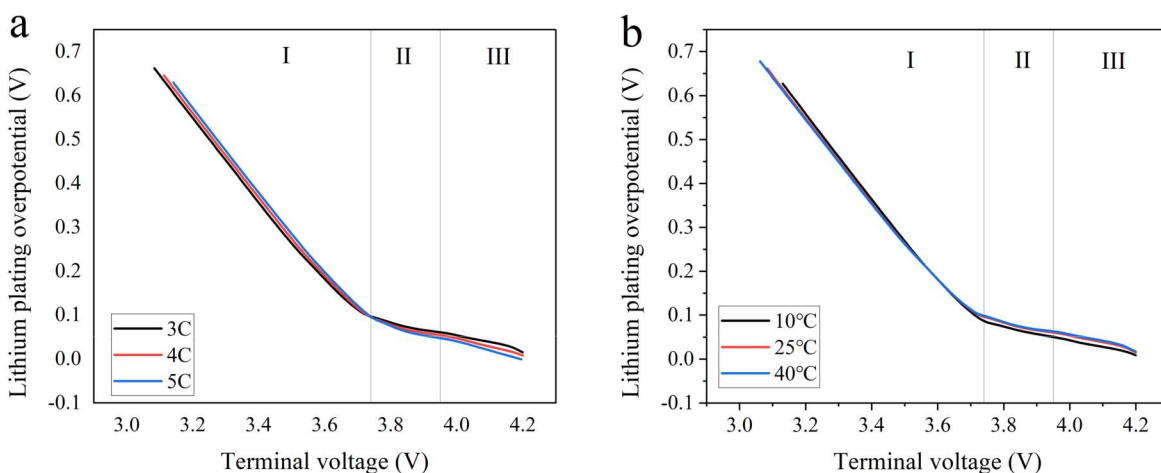


Figure 4. Correlation between lithium-plating overpotential and terminal voltage of cell at different operating temperatures and charging rates. (a) The correlation between the lithium-plating overpotential and terminal voltage of cell charged with 3 C, 4 C, and 5 C at 25 °C. (b) The correlation between the lithium-plating overpotential and terminal voltage of lithium-ion plating on cell charged at 3 C at 10 °C, 25 °C, and 40 °C.

demonstrates that the lithium-plating overpotential exhibits a three-stage voltage-dependent pattern during constant-current charging:

1. Stage I (Low voltage): approximately linear decrease in lithium-plating overpotential. Corresponding to the terminal voltage range from charging to 3.71 V.
2. Stage II (Medium voltage): near-plateau region with minimal decline. Corresponding terminal voltage within the range of 3.71 V~3.95 V
3. Stage III (High voltage): accelerated decrease as voltage approaches cutoff. Corresponding terminal voltage within the range of 3.95 V~4.2 V

Importantly, the voltage-overpotential correlation is fundamentally governed by intrinsic electrode kinetics rather than external current values. Variations in charging rates cause slight deviations in the terminal voltage at the transition points. The occurrence of the deviation indicates a change in the polarization of the cell anode and the entire cell.²³ The terminal voltage at the transition between the first and second stages is $3.71 \text{ V} \pm 0.02 \text{ V}$, while that between the second and third stages is $3.95 \text{ V} \pm 0.01 \text{ V}$. These findings demonstrate that variations in charging rates exert minimal influence on the correlation between lithium-plating overpotential and terminal voltage in the cell.

The fully discharged cell was charged to 4.2 V at a constant charging rate of 3 C under isothermal conditions of 10 °C, 25 °C, and 40 °C, respectively. Figure 4b shows that this three-stage pattern remains robust across temperatures. Only at lower operating temperatures is a slight shift in terminal voltage observed at the transition between Stages I and II, accompanied by a modest decrease in lithium-plating overpotential. Moreover, during the low-voltage stage, the linear slope of lithium-plating overpotential undergoes minor alteration. This phenomenon arises due to the reduced electrochemical performance of the cell at low temperatures, leading to uneven lithium-ion distribution in the electrode during the low-voltage stage, thereby altering cell polarization.²⁵

In order to further illustrate the correlation between cell polarization voltage and lithium-plating overpotential, the results of the variation of the cell polarization voltage with terminal voltage during the constant current charging process of a fully discharged cell to 4.2 V at different operating temperatures and different rate currents are simulated, as shown in Fig. 5.

The results presented in Fig. 5 demonstrate that the rate of decline in lithium-plating overpotential is positively correlated with the extent of cell polarization. During the initial phase of charging, the anode concentration polarization is large, leading to an increase in the total polarization of the cell and a rapid decrease in the lithium-plating overpotential. As charging progresses, the cell transitions into

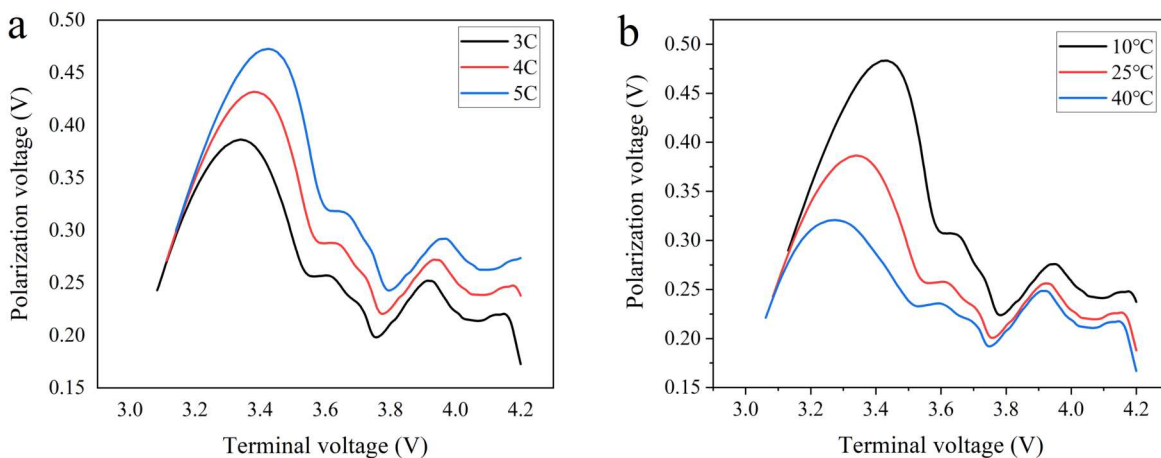


Figure 5. Correlation between polarization voltage and terminal voltage of cell at different operating temperatures and charging rates. (a) The correlation between the polarization voltage and terminal voltage of cell charged with 3 C, 4 C, and 5 C at 25 °C. (b) The correlation between the polarization voltage and terminal voltage of cell charged with 3 C at 10 °C, 25 °C, and 40 °C.

a more balanced operational state, resulting in reduced polarization and a deceleration in the decline of lithium-plating overpotential. As charging approaches completion, concentration polarization at the cathode gradually intensifies, elevating the overall polarization of the cell, and the lithium-plating overpotential accelerates to decrease again.²⁶ The average polarization voltage of the cell rises with increasing charging current or decreasing operating temperature. This phenomenon is mainly arises from the heightened lithium-ion concentration gradient within the active material particles of both the cathode and anode driven by higher charging currents or lower temperatures, leading to an increase in concentration polarization voltage at both electrodes. Additionally, this effect can also be attributed to the intensification of internal electrochemical reactions within the cell, which elevates the electrochemical polarization voltage.

Fast Charging Strategy for Suppressing Lithium Plating in Cells

Building on the full-range electrochemical correlation between terminal voltage and lithium-plating overpotential established in section “Basic Principle”, we develop a voltage-driven fast-charging strategy that minimizes the rate of decline in lithium-plating overpotential without compromising charging efficiency. The proposed strategy regulates current profiles solely based on terminal voltage measurements, eliminating the operational need for onboard electrochemical models, state estimators, or reference electrodes. The suppression of lithium plating is thus indirectly inferred from macroscopic electrochemical indicators—such as the observed improvement in capacity retention and significant reduction in polarization—rather than being directly monitored via in situ sensing during operation. Owing to the correlation between terminal voltage and lithium-plating overpotential, the charging strategy is extrapolated as a function of charging current variation controlled by terminal voltage.

The charging process is divided into three different stages summarized in section “Establishment of full-range voltage–overpotential correlation”. To mitigate the adverse effects of cell polarization voltage during the low voltage stage, the initial charging current should be kept at a relatively low level. As the terminal voltage rises and polarization decreases, the charging current that the cell can accept gradually increases. Hence, during the low voltage stage, the charging begins at 0 A, with the current increasing linearly over time. Upon reaching 3.71 V, the cell voltage enters the medium-voltage stage, and the charging current rises to its maximum value. The corresponding linear current ramp is calculated by the model. In this stage, the lithium-plating overpotential decreases at the slowest rate. At this time, the cell is capable of sustaining constant current charging at its maximum allowable rate until the terminal voltage

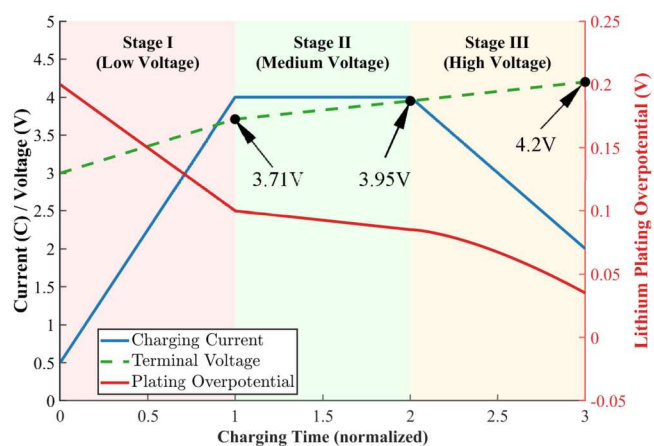


Figure 6. Schematic illustration of the proposed fast-charging strategy.

reaches 3.95 V. Entering the high voltage stage, the lithium-plating overpotential accelerates to decrease again. At this point, the charging current decreases linearly with time until charging is completed. The schematic illustration of fast charging strategy is shown in Fig. 6. Based on the lithium-plating overpotential characteristics, the transition voltages between stages have been identified. Therefore, this charging strategy only needs to determine two parameters: the maximum constant current C_{MAX} in the medium voltage stage and the current drop rate C_D in the high voltage stage.

Determination of current parameters.—The specific parameters of the current curve need to be determined when applying the fast charging strategy in practice. In this section, numerical analyses are conducted to investigate the impact of C_{MAX} and the C_D on the fast charging performance, to culminating in the determination of the optimal charging current curve. Firstly, numerical studies are conducted on the variation of lithium-plating overpotential at the anode–separator interface during constant current charging rates ranging from 3 C to 8 C at 25 °C. The results, as illustrated in Fig. 7, demonstrate that when the charging rate exceeds 6 C, the lithium-plating overpotential at the anode–separator interface drops below 0 V, indicating a heightened risk of lithium plating at the anode under high-rate conditions. The effect of varying the C_{MAX} on fast charging performance, while maintaining a constant current decrease slope, is analyzed. The corresponding results are shown in Fig. 8.

Figure 8a illustrates the effect of C_{MAX} on the lithium-plating overpotential at the anode. The results reveal that increasing C_{MAX} significantly reduced the lithium plating overpotential, thereby

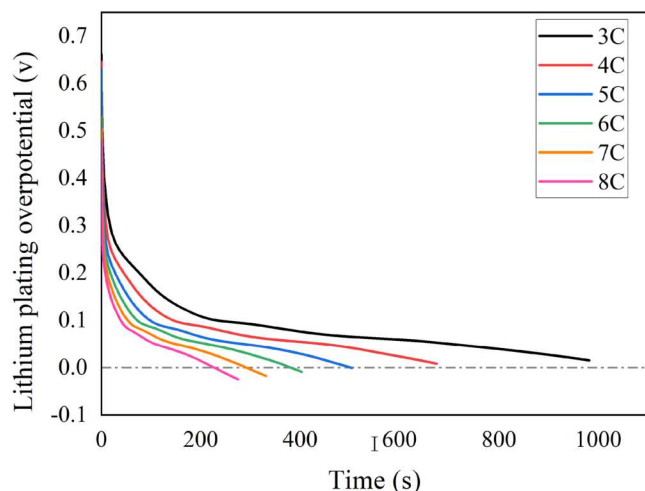


Figure 7. Overpotential values of lithium plating at the interface between the anode and separator when charging the cell at different rates.

elevating degradation risks. Furthermore, the C_{MAX} determines the minimum value of lithium-plating overpotential throughout the entire charging process.

Figure 8b illustrates the effect of C_{MAX} at different rates on the charging efficiency. While variations in C_{MAX} had a negligible effect on the cutoff SOC, they markedly influenced total charging time and overall efficiency. Higher C_{MAX} values enhanced efficiency but necessitated stricter safety constraints.

The charging current during the low and high voltage stages suppresses lithium plating reactions in the cell. Therefore, we believe that the critical C_{MAX} for the cell to not occur lithium plating during medium voltage stage charging is 6 C.

Afterwards, fix the C_{MAX} to 6 C and analyze the impact of changes in the rate of current decrease on the charging effect. The results are shown in Fig. 9.

Figures 9a and 9b illustrate the lithium-plating overpotential curves during cell charging at various C_D , along with the corresponding SOC values at the charging cutoff. The C_D significantly influences the trend of anodic lithium-plating overpotential during high voltage charging of the cell. When the current decreases rapidly, i.e. $C_D = -1.8 \text{ mA s}^{-1}$, in the later stage of charging, the lithium-plating overpotential rebounds far away from the threshold potential of lithium plating. This behavior leads to additional and unnecessary energy input during charging. For electric vehicle applications, fast

charging is typically required only up to 80 SOC. The accelerated aging and safety issues associated with fast charging in the 80% to 100% SOC range have increased significantly.²⁷ Therefore, fast charging beyond 80% SOC is not considered in this study. Meanwhile, at high states of charge, the cell exhibits low charging efficiency, indicating that the current strategy may be overly conservative. When the rate of current decrease is small, i.e. $C_D = -1.4 \text{ mA s}^{-1}$, the lithium-plating overpotential accelerates its decline in the later stages of charging, thereby elevating the risk of lithium plating reactions. A relatively moderate C_D should be adopted to satisfy fast charging requirements.

References 28, 29 indicate that the current and polarization along the direction of the electrode current collector foil are non-uniform. Therefore, a buffer value needs to be reserved for the lithium-plating overpotential threshold. A buffer value of 10 mV has been selected, indicating that a lithium plating reaction will occur when the cell's lithium-plating overpotential falls below this threshold.¹⁰ Building upon the findings from the previous section, and in order to meet the requirements of the lithium-plating overpotential threshold for lithium plating in cell, the maximum charging current $C_{MAX} = 6 \text{ C}$ and the current drop rate $C_D = -1.6 \text{ mA s}^{-1}$ are selected as the charging current parameters for the fast charging strategy in the cell to suppress lithium plating. The resulting protocol operates as follows:

Stage I: Current increases linearly from zero until the terminal voltage reaches 3.71 V.

Stage II: Charging proceeds at a constant C_{MAX} of 6 C until the voltage attains 3.95 V.

Stage III: The current ramps down following the C_D profile of -1.6 mA s^{-1} until completion.

The model computes the current rise profile by linearly increasing the current with time and mapping the corresponding cell voltage. This design ensures that the applied current is continuously constrained and controlled by the cell voltage. Figure 10 presents an example of a fast charging strategy. Figure 10a illustrates the distribution of charging current and cell voltage throughout the charging process, while Fig. 10b depicts the relationship between charging current and cell terminal voltage, reflecting how terminal voltage constrain the allowable charging current.

The aforementioned numerical analysis results illustrate the influence of charging current parameters on the effectiveness of various charging methods, as well as a methodology for current parameter screening. While the ETA model was utilized for offline parameter screening, the final strategy requires only real-time terminal voltage for implementation, ensuring compatibility with standard battery management systems (BMS).

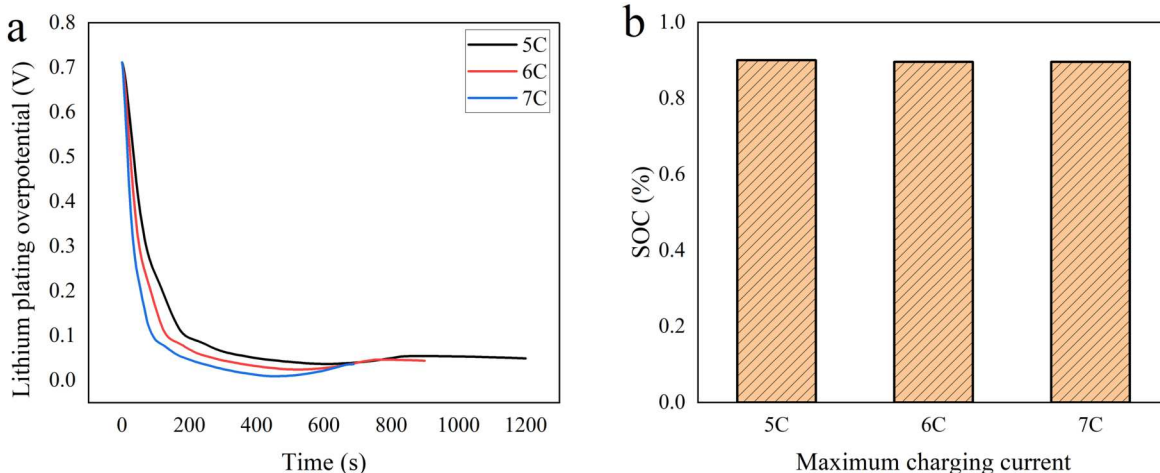


Figure 8. The effect of different C_{MAX} on fast charging performance. (a) The lithium-plating overpotential change curve during the charging process. (b) Cell charging cut-off SOC.

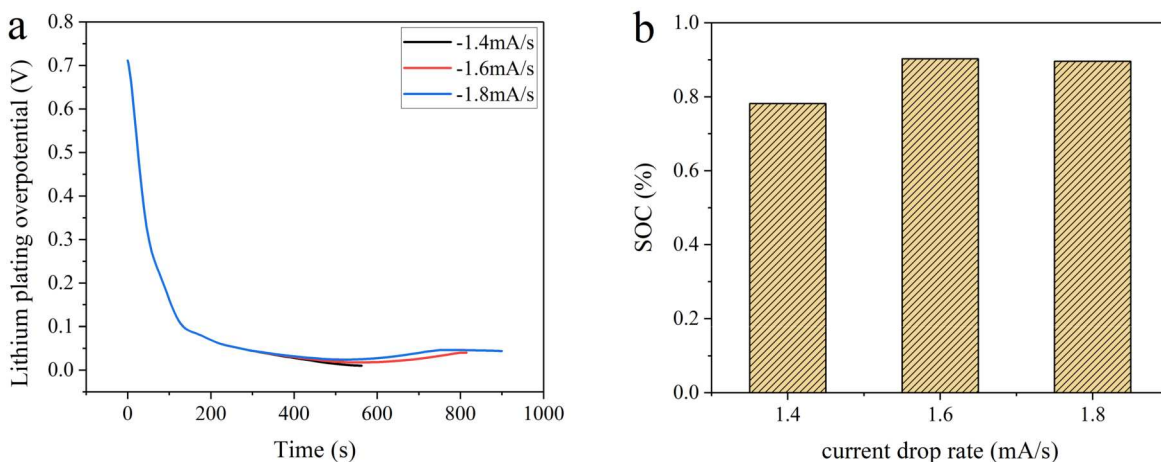


Figure 9. The impact of different C_D on the fast charging effect. (a) The variation curve of lithium-plating overpotential. (b) Cell charging cut-off SOC.

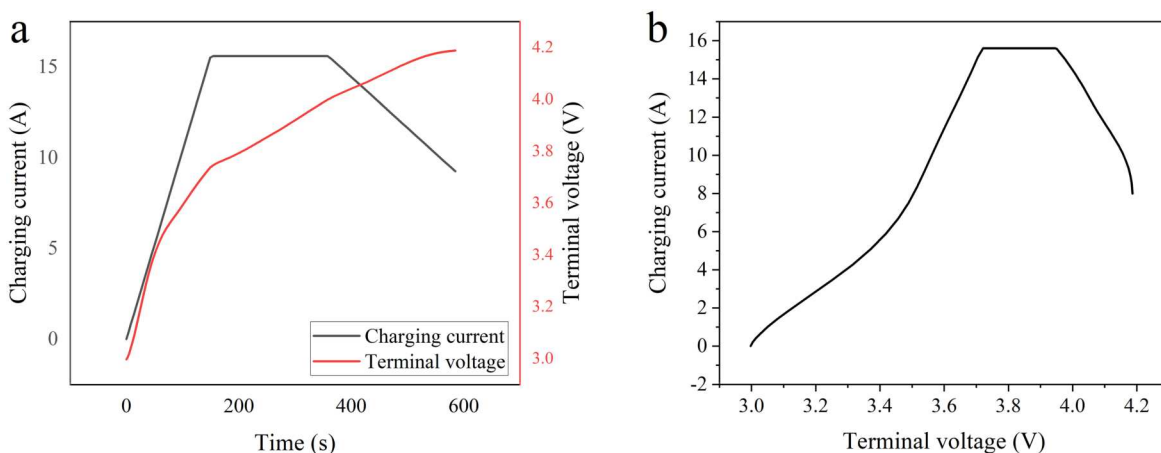


Figure 10. Example of fast charging strategy in cell to suppress lithium plating. (a) Charging current and voltage distribution. (b) The relationship between charging current and voltage.

Simulation testing of fast charging strategy.—This section presents a comprehensive parameterized sweeping analysis of the current parameters for the fast charging strategy to suppress lithium plating and evaluates its effectiveness relative to the conventional CC-CV charging approach. A parameterized swapping analysis was performed on the combination of current parameters, with the C_{MAX} ranging from 5 C to 7 C and the C_D spanning from -1.6 mA s^{-1} to -2.2 mA s^{-1} . For comparison, conventional CC-CV charging is also simulated across a range of constant-current rates from 2 C to 7 C. All simulations charged a fully discharged cell to 80% SOC. The results of charging efficiency and cell safety are presented in Fig. 11. The data points of the fast charging strategy to suppress lithium plating are all located to the left of the data points for the CC-CV method. This suggests that, under identical charging efficiency, the proposed strategy raises the lithium-plating overpotential, thereby reducing the risk of lithium plating. Equivalently, for the same safety margin, the proposed strategy shortens the charging time, reflecting a more favorable safety-efficiency trade-off. This systematic analysis proves that the charging profile derived from electrochemical mechanism outperforms CC-CV across the entire parameter space, not only at the optimized parameter set. This further validates the mechanistic correctness and generality of the voltage-overpotential correlation established in section “Basic Principle”.

To further evaluate the effectiveness of the fast charging strategy in suppressing lithium plating, we compared the fast charging strategy to suppress lithium plating, as presented in the “Determination of current parameters” section, and against an

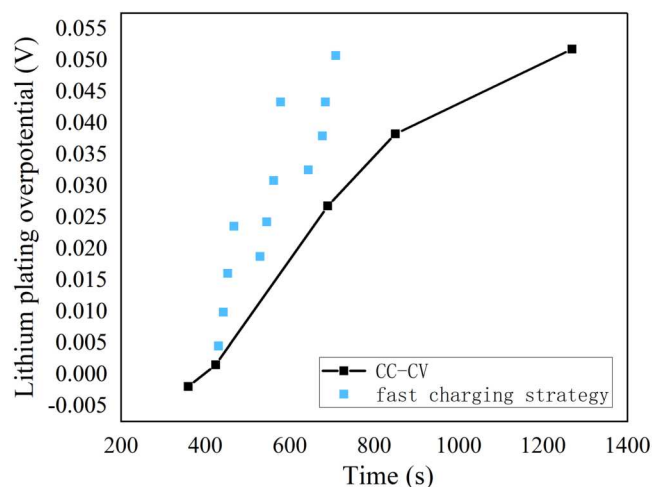


Figure 11. The results of the parameterized swapping study on fully discharged cell charged to 80% SOC using the fast charging strategy to suppress lithium plating and a CC-CV charging strategy.

equivalent CC-CV strategy with the same charging efficiency. An equivalent charging strategy is defined as one that delivers the same charging duration and total input energy as the proposed fast-charging strategy. The charging process is simulated based on the

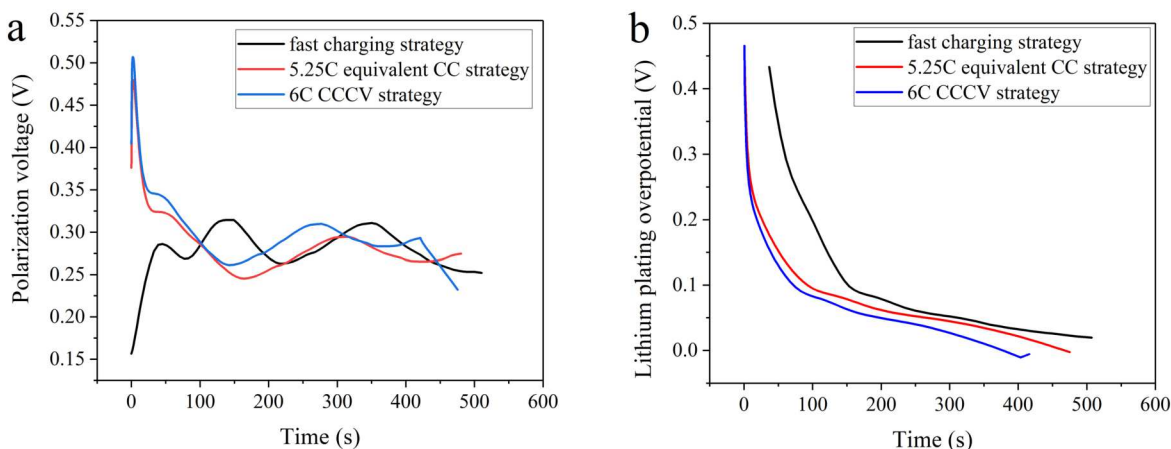


Figure 12. The results of a single charge-discharge cycle using the fast charging strategy to suppress lithium plating or CC-CV method in the cell. (a) Polarization voltage during charging process. (b) Lithium-plating overpotential during charging process.

model developed in section “Basic Principle”. According to model calculations, the equivalent constant current is 31.5 A (5.25 C). Additionally, the results of charging the cell using a 6 C CC-CV strategy were also compared. The polarization voltage and lithium-plating overpotential during the cell charging process are presented in Fig. 12.

Figure 12a illustrates the polarization voltage profiles of the cell charged using the fast charging strategy to suppress lithium plating, equivalent CC-CV charging strategy, and 6 C CC-CV charging strategy, respectively. The results indicate that compared to the equivalent CC-CV charging strategy, the fast charging strategy to suppress lithium plating can effectively suppress polarization in low-SOC region. The cell exhibits an average polarization voltage that is 45 mV lower than that of the equivalent CC-CV strategy and 55 mV lower than the 6 C CC-CV strategy throughout the charging process. These findings confirm that the proposed fast charging strategy to suppress lithium plating more effectively mitigates cell polarization during high-rate charging.

Figure 12b presents the lithium-plating overpotential profiles at the anode-separator interface for cell of the interface charged using the fast charging strategy to suppress lithium plating, equivalent constant current charging strategy, and 6 C CC-CV charging strategy. The results demonstrate that the decline in lithium-plating overpotential is more gradual during the initial charging phase, implying that additional safety margins are preserved, allowing for higher current rates in subsequent charging stages. When the lithium plating suppress fast charging strategy reaches its maximum charging current, even if the charging current is higher at this time, the

lithium-plating overpotential value is still higher than that of using the equivalent CC-CV charging strategy, with even the smallest difference being 4 mV. At the end of charging, the lithium-plating overpotential difference expands to 12 mV. The average lithium-plating overpotential value throughout the entire charging process is 13% higher. The lithium-plating overpotential of the cell, when charged using the fast charging strategy to suppress lithium plating, is significantly higher than that under the 6 C CC-CV protocol, with the minimum difference reaching 11 mV. These findings suggest that, at equivalent charging efficiencies, the proposed fast charging strategy can effectively reduce the risk of lithium plating.

Two multistage constant-current charging strategies and one pulse-charging strategy were constructed for comparison with the proposed lithium plating-suppression fast-charging strategy, as illustrated in Fig. 13. Figure 13a compares the charging profiles of the investigated strategies. MCC-A adopts a four-stage constant-current scheme designed to emulate the characteristics of the lithium plating-suppression fast-charging strategy, in which the current ramp-up and ramp-down processes are approximated by stepwise current levels with a step size of 2 C and identical duration at each stage. MCC-B employs a six-stage constant-current charging scheme, where charging is initiated at 6 C and the current is subsequently reduced in uniform 1 C increments, with equal duration maintained at each stage. The timing of the initial current reduction in MCC-B is deliberately aligned with that of the proposed fast-charging strategy to ensure direct comparability. For the pulse-charging strategy, the charging and discharging rates are set to 6 C and 3 C, respectively, with a duty cycle of 0.98. The evolution of lithium-plating overpotential during charging

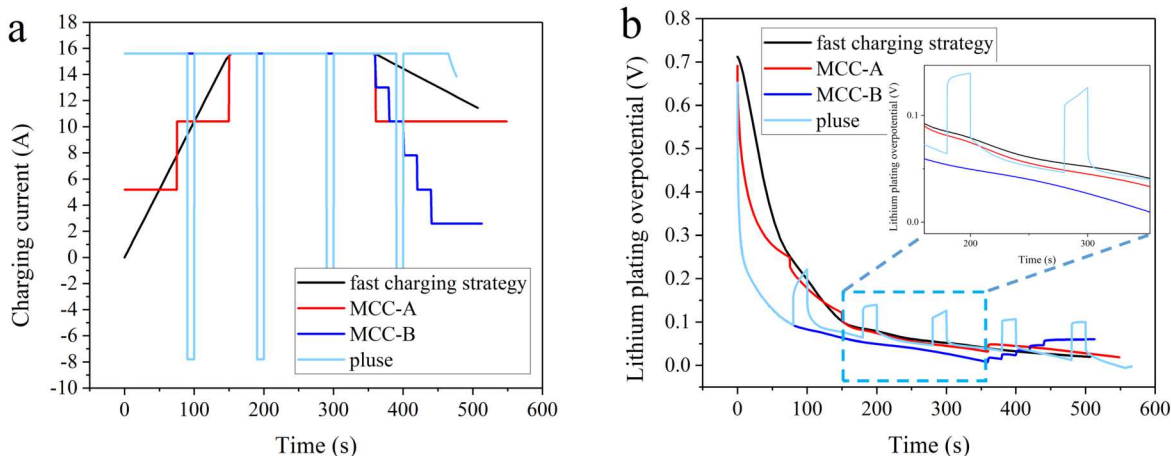


Figure 13. Comparison of fast charging strategy to suppress lithium plating with other advanced fast charging strategies. (a) Current profile. (b) Lithium-plating overpotential.

Table I. Statistical summary of lithium plating overpotential under different charging strategies.

Charging strategy	Average value(mV)	Minimum value (mV)
MCC-A	214.2	18.9
MCC-B	145.4	7.1
Fast charging strategy	247.8	19.5
Pulse charging strategy	80.3	-5.9

is presented in Fig. 13b. Similarly, the lithium plating-suppression fast-charging strategy exhibits a more gradual decline in lithium-plating overpotential, remaining 5.3 mV higher than that of the adjacent MCC-A strategy even at the maximum charging current. Statistical results of the average and minimum lithium-plating overpotentials during charging are summarized in Table I. When the proposed fast-charging strategy is applied, both the average and minimum lithium-plating overpotentials remain higher, thereby preserving a larger safety margin throughout charging.

Experimental verification of fast charging strategy.—This section validates the efficacy of the fast charging strategy to suppress lithium plating in vehicle lithium-ion cell through experiments. The verification experiment employed the same battery and experimental setup as described in section “Basic Principle”. The cell is housed in a Kingjo LK-H80G environmental chamber constant temperature and humidity box, set the constant temperature to the recommended test temperature of 23 °C by the cell manufacturer. This arrangement ensures a stable thermal environment, thereby mitigating the potential impact of temperature fluctuations on the experimental results.

The cell is subjected to cyclic durability testing using the fast charging strategy to suppress lithium plating and an equivalent CC-CV charging strategy. The manufacturer specifies a maximum charging current of 6 A (2.3 C) for this cell. Therefore, the profile of the fast charging strategy to suppress lithium plating follows the parameters determined in section “Determination of current parameters”: The charging current increases linearly with time starting from 0 A to 7.8 A, when the cell voltage reaches 3.71 V. A maximum current of 7.8 A (3 C) is maintained within the 3.71–3.95 V window. The current subsequently ramped down at a rate of -1.6 mA s^{-1} until the SOC until the 80% SOC. The cell was then discharged to 2.5 V at 5.2 A (2 C) under a constant-current (CC) protocol. A 5-minute rest period is imposed between the charging and discharging steps. Based on model calculations, the corresponding equivalent CC-CV charging current is determined to be 6 A. The equivalent charging cycle proceeds as follows: the cell is charged with a constant current of 6 A until reaching a cut-off voltage of 4.2 V, then transitioned to a constant-voltage phase until 80% SOC. Subsequently, the cell is discharged to 2.5 V at a current of 5.2 A (2 C). Similarly a 5-minute rest period is applied between steps. The charge retention rate (CR) of the cell is recorded after each cycle. A standard capacity test (SCT) is conducted at intervals of 25 cycles. The SCT procedure consists of charging the cell using the CC-CV method at 2.6 A (1 C). Once the cell voltage reaches the 4.2 V cut-off voltage, the system switches to constant voltage charging until the current decreases to 0.26 A (0.1 C). After a 30-minute rest, a 1 C discharge is executed to determine the reference discharge capacity. The fast-charging strategy to suppress lithium plating and the equivalent CC-CV strategy are evaluated at 25 °C and 40 °C, respectively, with the results presented in Fig. 14.

Figure 14a illustrates the charge retention rate results of 200 cycle tests on two sets of cell at 25 °C. The charge retention rate of a cell

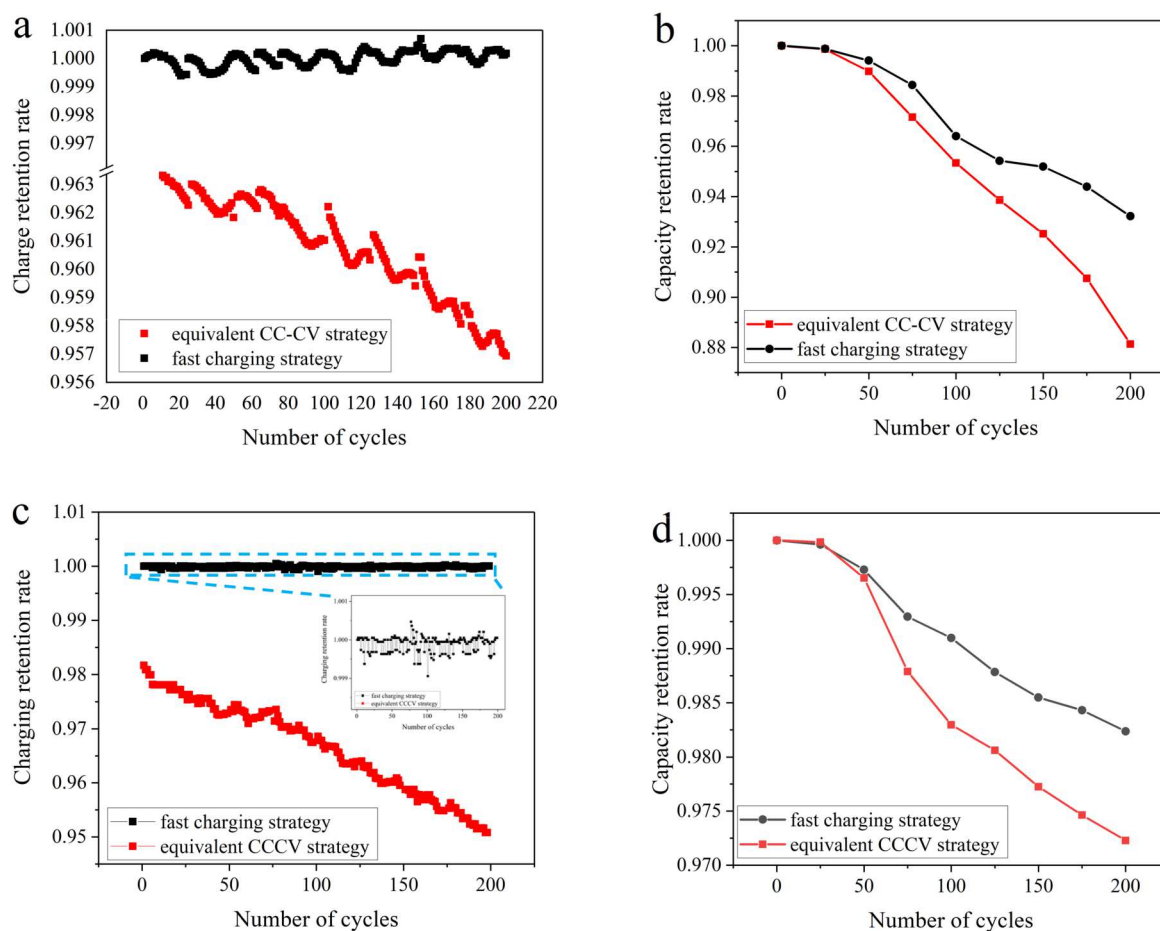


Figure 14. Cell cycle durability test. (a) Charge retention rate at 25 °C. (b) Capacity retention rate at 25 °C. (c) Charge retention rate at 40 °C. (b) Capacity retention rate at 40 °C.

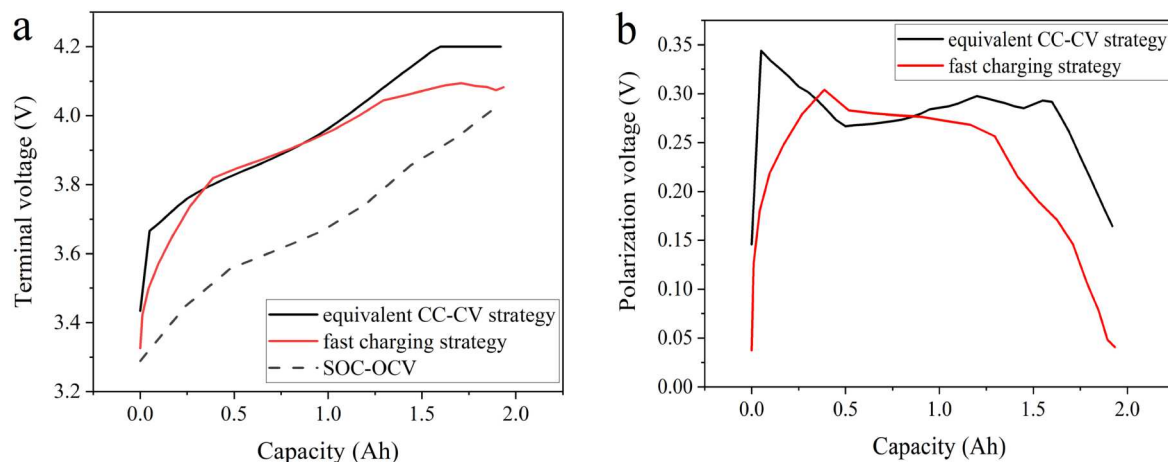


Figure 15. Polarization during the cell charging process. (a) Changes in voltage and capacity. (b) Polarization voltage.

charged using the fast charging strategy to suppress lithium plating fluctuates around 100%, indicating robust charge replenishment capability and with no observable increase in cell polarization. In contrast, the charge retention rate of the cell charged using the equivalent CC-CV method exhibits continuous fluctuations and a decline, suggesting that the lithium plating reaction during charging leads to the increased cell polarization.¹⁰ The results in Fig. 14b indicate that, throughout 200 charge-discharge cycles, the capacity retention rate of the cell charged with the fast charging strategy to suppress lithium plating consistently outperforms that of the cell charged using the equivalent CC-CV method, achieving an additional 6.8% capacity retention after 200 cycles. During the experiments, confounding factors such as charging time and temperature—which could trigger additional side reactions—are strictly controlled. These results indicate that the extra capacity loss primarily originates from lithium plating. Compared with the CC-CV charging strategy, the capacity loss associated with lithium plating was reduced by approximately 50% after 200 cycles, confirming under the equivalent charging efficiency, the effectiveness of the proposed fast charging strategy in mitigating lithium plating-induced degradation. The results shown in Figs. 14c and 14d indicate that the fast-charging strategy yields comparable charge and capacity retention at 40 °C and 25 °C. The cell charged using the fast-charging strategy exhibits an additional 1.3% capacity retention after 200 cycles. The reduced disparity in capacity retention at elevated temperatures can be attributed to enhanced electrode kinetics, which partially mitigates capacity loss.

We compared the polarization degree of the cell under two charging methods. Figure 15a comparison of terminal voltage against the calibrated OCV curve reveals that under conditions of delivering the same capacity to the cell, the terminal voltage of the cell utilizing the fast charging strategy to suppress lithium plating is mostly lower than when using the CC-CV charging strategy. Macroscopically, this phenomenon reflects a reduction in cell polarization and an enhancement in charging efficiency. To provide a clearer representation of the cell's polarization, Fig. 15b presents the polarization voltage under the fast-charging and CC-CV strategies. Throughout the charging process, the average polarization voltage of the cell charged under the fast-charging strategy is 37% lower than that of the cell charged by the CC-CV method. These results aligning with the mechanistic prediction that suppressing early-stage polarization slows the decline of lithium-plating overpotential.

Conclusions

This study establishes a mechanistic foundation for voltage-driven fast charging by revealing a universal correlation between terminal voltage and lithium-plating overpotential across the full

charging window. Key mechanistic insights reveal that polarization voltage peaks early in charging (low SOC) and then gradually declines—a behavior driven by incomplete lithium-ion diffusion, which creates large concentration gradients in anode particles, intensifying both anode concentration polarization and overall cell polarization. This enhanced polarization directly accelerates the drop in lithium-plating overpotential. Further analysis identifies a stage-dependent relationship between terminal voltage and lithium-plating overpotential: an approximately linear correlation at low voltages, a slowed decay rate at medium voltages, and a renewed rapid decrease at high voltages. Notably, these characteristics remain robust across varying charging rates and temperatures.

Guided by these findings, a three-stage charging protocol is designed: (1) current ramps up from zero under voltage constraints until reaching the maximum safe rate; (2) charging continues at this constant maximum current through the medium-voltage stage; and (3) current decreases linearly during the high-voltage stage to mitigate plating risk. This fast-charging strategy that modulates current in accordance with voltage-dependent overpotential evolution, requiring no electrochemical models, state estimators, or reference electrodes during operation. Parametric analysis confirms that the maximum charging current governs the minimum lithium-plating overpotential, while the current decay rate in the final stage controls its late-stage trend.

Extensive parameter sweeping shows that the proposed method consistently outperforms CC-CV charging across the entire parameter space, achieving a superior balance between safety and charging speed. Compared with conventional CC-CV charging operated at the same energy efficiency, the proposed strategy raises lithium-plating overpotential by 13% under typical conditions. Compared with other advanced MCC and pulse charging strategies, the lithium plating overpotential is increased by 13.5%–67.4%, effectively reducing deposition risk. Experiments conducted on commercial 18650 NMC811 cells show that the strategy enhances capacity retention by maximum of 6.8% after 200 cycles and reduces polarization voltage by 37%, thereby simultaneously mitigating lithium plating and polarization.

Although the proposed strategy effectively suppresses lithium plating in fresh cells under room and elevated temperatures, its long-term robustness and adaptability to extreme operating conditions require further systematic investigation. At low temperatures, severely hindered ionic transport kinetics and reduced usable capacity may necessitate temperature-adaptive modulation of the voltage thresholds. Furthermore, as the cell ages, degradation-induced phenomena—lithium inventory loss and active material degradation—may modify the intrinsic correlation between terminal voltage and plating overpotential. Future work will therefore focus on developing an adaptive control framework that integrates thermal sensitivity and aging-induced state-of-health evolution, thereby

enabling precise suppression of lithium plating across the full operational envelope and throughout the battery lifecycle. In future work, post-mortem analysis will be conducted on the tested cells to systematically characterize the spatial distribution and morphology of lithium plating on the electrode surfaces. In addition, differential voltage analysis and incremental capacity analysis will be employed to quantitatively evaluate the extent of lithium plating.

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CRediT Authorship Contribution Statement

Zheng Wang: Conceptualization, Writing—original draft, Validation. **Xiaolan Wu:** Conceptualization, Project administration, Supervision, Resources, Funding acquisition, Writing—review & editing. **Naixing Yang:** Validation. **Huijie Shi:** Validation. **Oluwatunmise Sharon Banjoko:** Writing—review & editing. **Zeyu Chen:** Writing—review & editing, Funding acquisition, Project administration.

Appendix A. Model Parameterization Model Parameterization

Table A-I. Electrochemical thermal aging coupling model parameter.

Parameter	Anode	Separator	Cathode
Electrode thickness	56 μm^{30}	16 μm^{30}	50 μm^{30}
Particle radius	14.75 μm^{30}		8.1 μm^{30}
Solid phase porosity	0.56 ³⁰		0.685 ²²
Liquid phase porosity	0.268 ³⁰	0.55 ³⁰	0.306 ³⁰
Maximum concentration of lithium in the active material	31507 mol m ^{-3a)}		38271 mol m ^{-3a)}
Initial concentration of lithium ions in the liquid phase		1000 mol m ^{-3a)}	
Solid state diffusivity	3.9*10 ⁻¹³ m ² /s ³⁰		1.3*10 ⁻¹³ m ² /s ³⁰
Liquid state diffusivity		2.7877*10 ⁻¹⁰ m ² /s ³¹	
Major reaction exchange current density	2.1 A m ⁻²³²		2 A m ⁻²³³
Lithium plating exchange current density	10 A m ⁻²³⁴		
SEI exchange current density	1.6*10 ⁻¹⁶ A m ⁻²²²		
Conductivity of active material	100 S m ⁻¹³⁰		0.17 S m ^{-1a)}
Density	1400 kg m ⁻³	492 kg m ⁻³³⁰	3400 kg m ^{-3a)}
Specific heat capacity	1095.4 J kg ⁻¹ K ⁻¹³⁰	1978 J kg ⁻¹ K ^{-1a)}	1016.6 J kg ⁻¹ K ⁻¹³⁰
Anode equilibrium potential	$E_{\text{ocv,n}} = 9.599 \times 10^{10} \exp(-(x+0.7041)/0.1426^2) + 0.1276 \exp(-(x-0.04364)/0.09979^2) + 1.641 \exp(((x+5.073)/3.308^2)^a)$		
Cathode equilibrium potential	$E_{\text{ocv,p}} = -1220 \times 7 + 5085 \times 6 - 8801 \times 5 + 8172 \times 4 - 4379 \times 3 + 1348 \times 2 - 221.3x + 19.17^a)$		

a) adjust and assumed values.

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